

智能仪器设计基础（一）

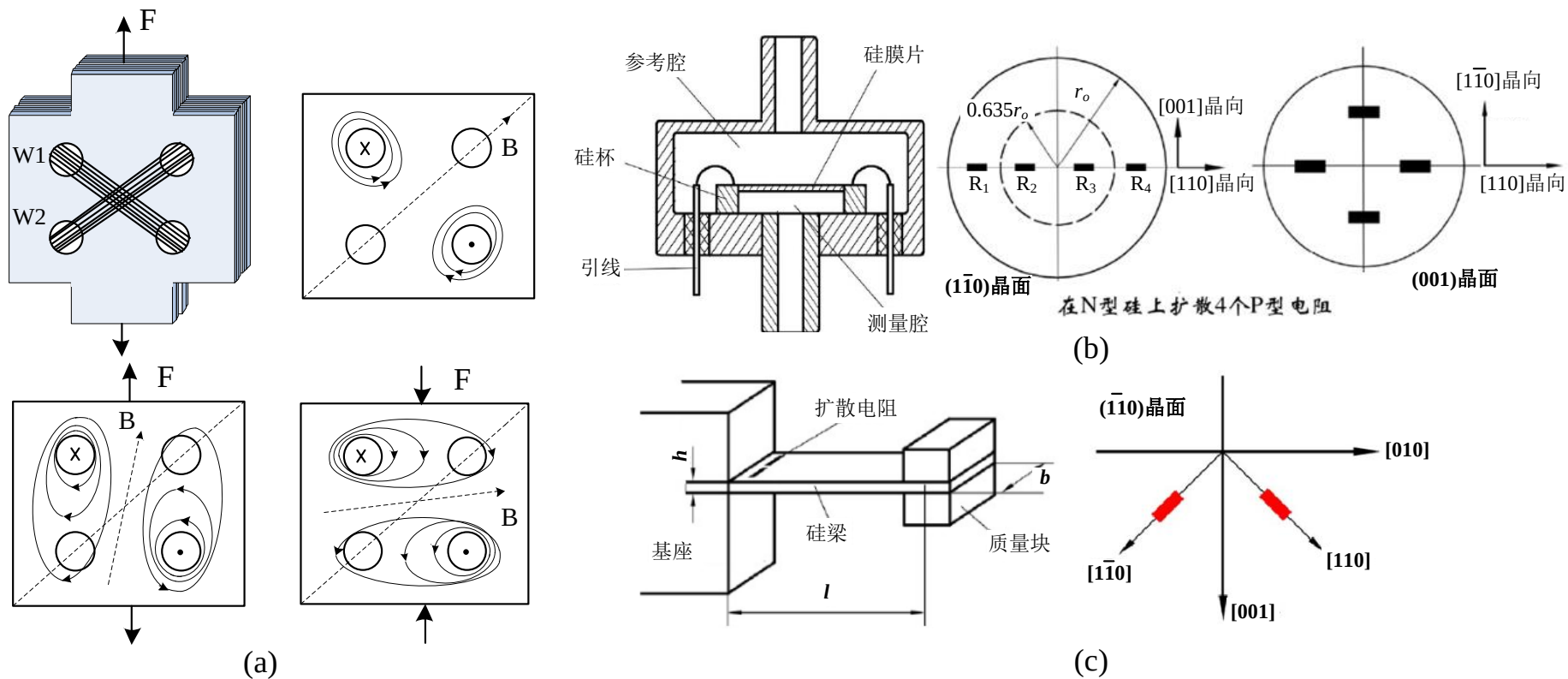
西安交通大学

主讲人：曾翔君



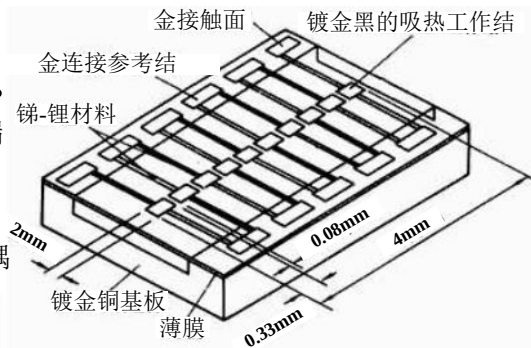
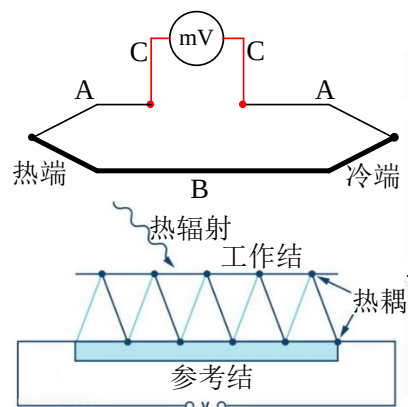
02

传感器的分类和原理 (下)

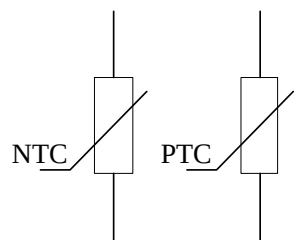


压磁和压阻效应传感器：(a) 压磁效应；(b) 压阻效应—压力传感器；(c) 压阻效应—加速度传感器

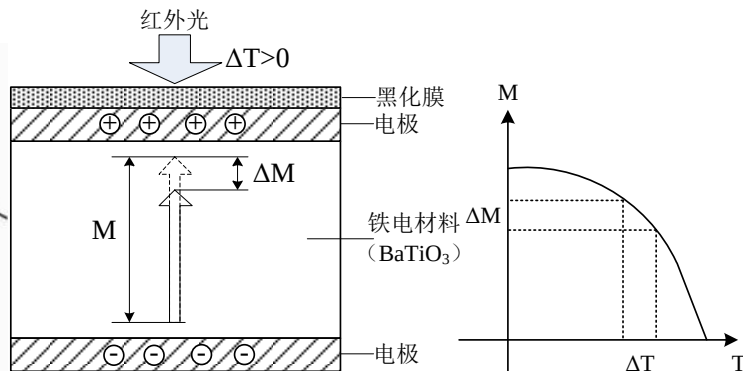
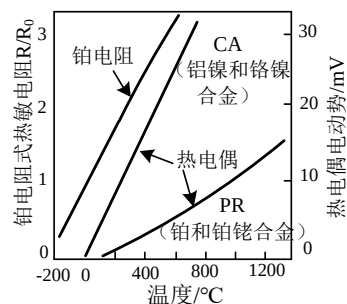
物性型传感器—热敏效应传感器



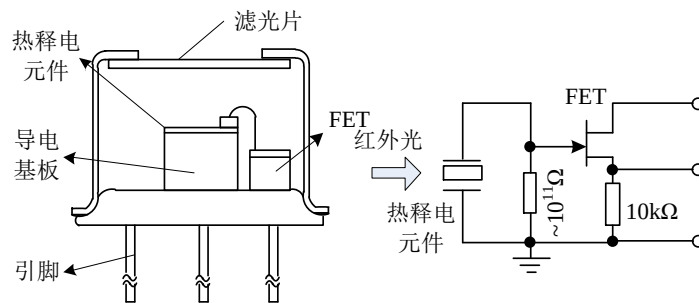
(a)



(b)

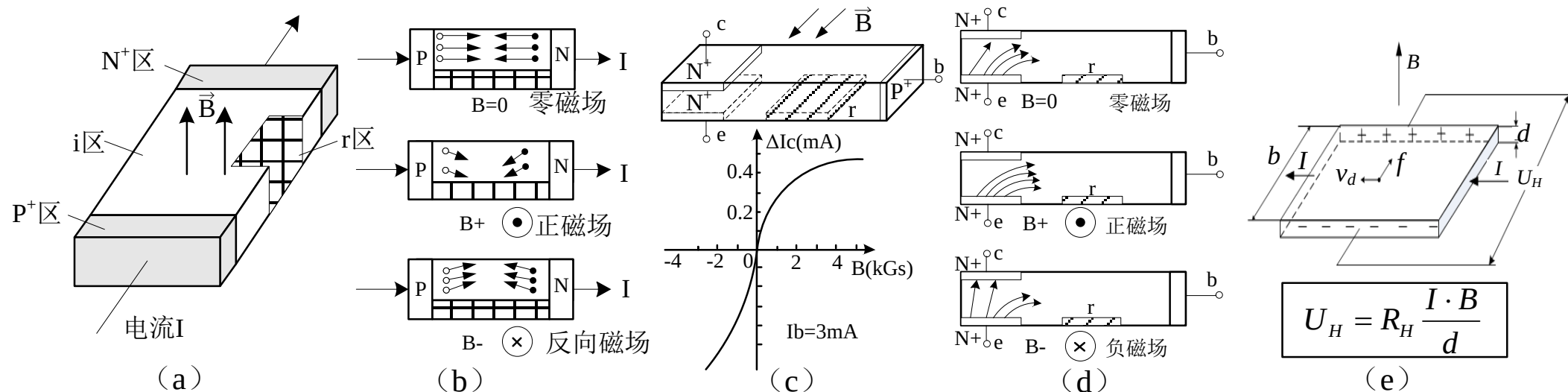


(c)

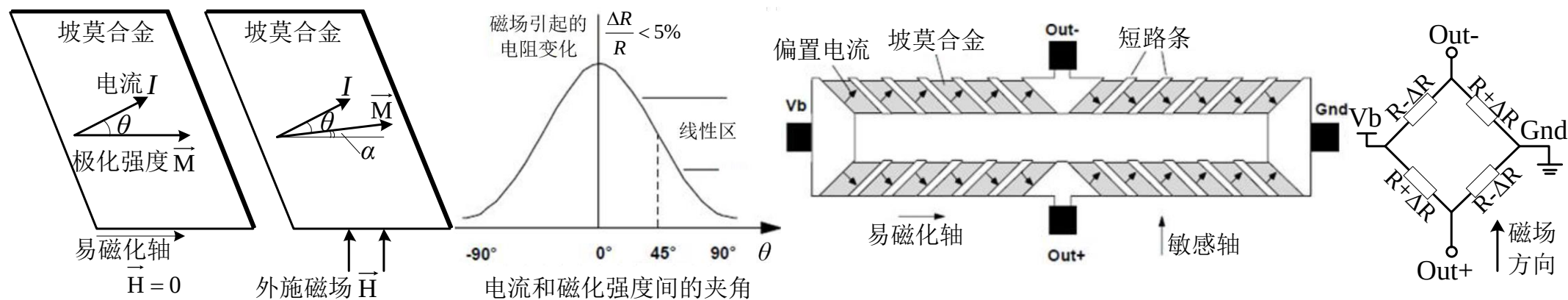


(d)

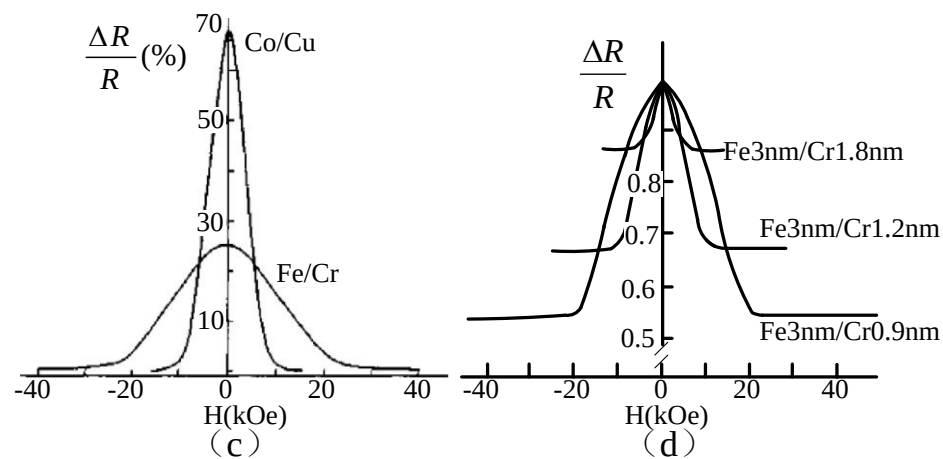
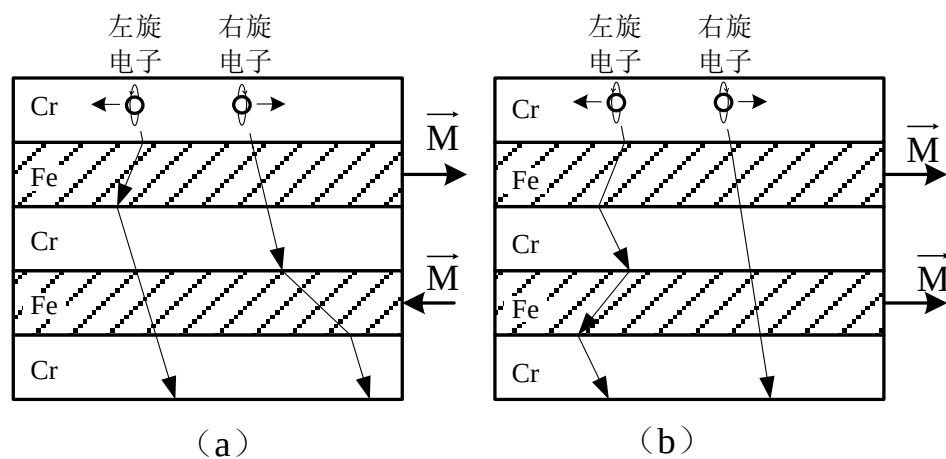
热敏特性及热敏传感器：(a)热电偶和热电堆；(b)热敏电阻；(c)热释电效应；(d)热释电传感器



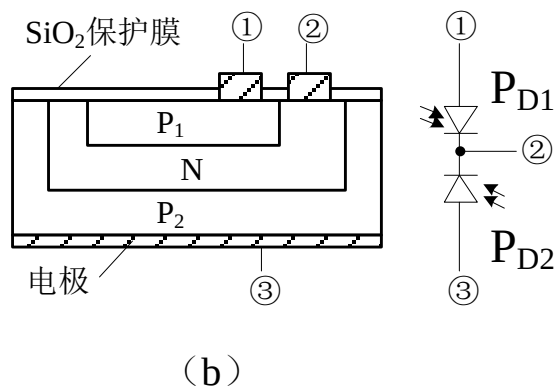
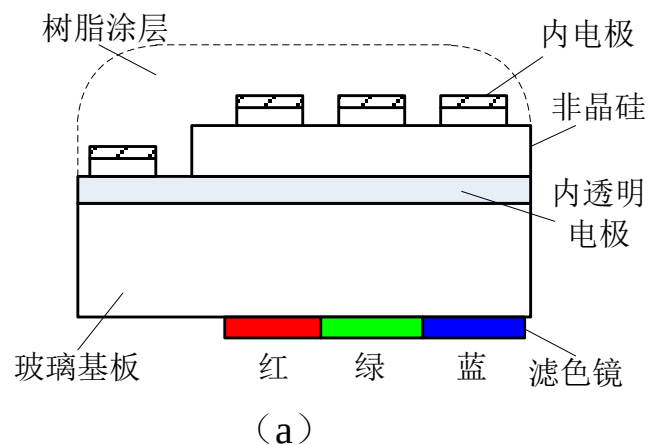
(a)磁敏二极管结构；(b)磁敏二极管原理；(c)磁敏三极管结构和典型特性；(d)磁敏三极管原理；(e)霍尔效应



各向异性 (AMR) 磁阻效应传感器



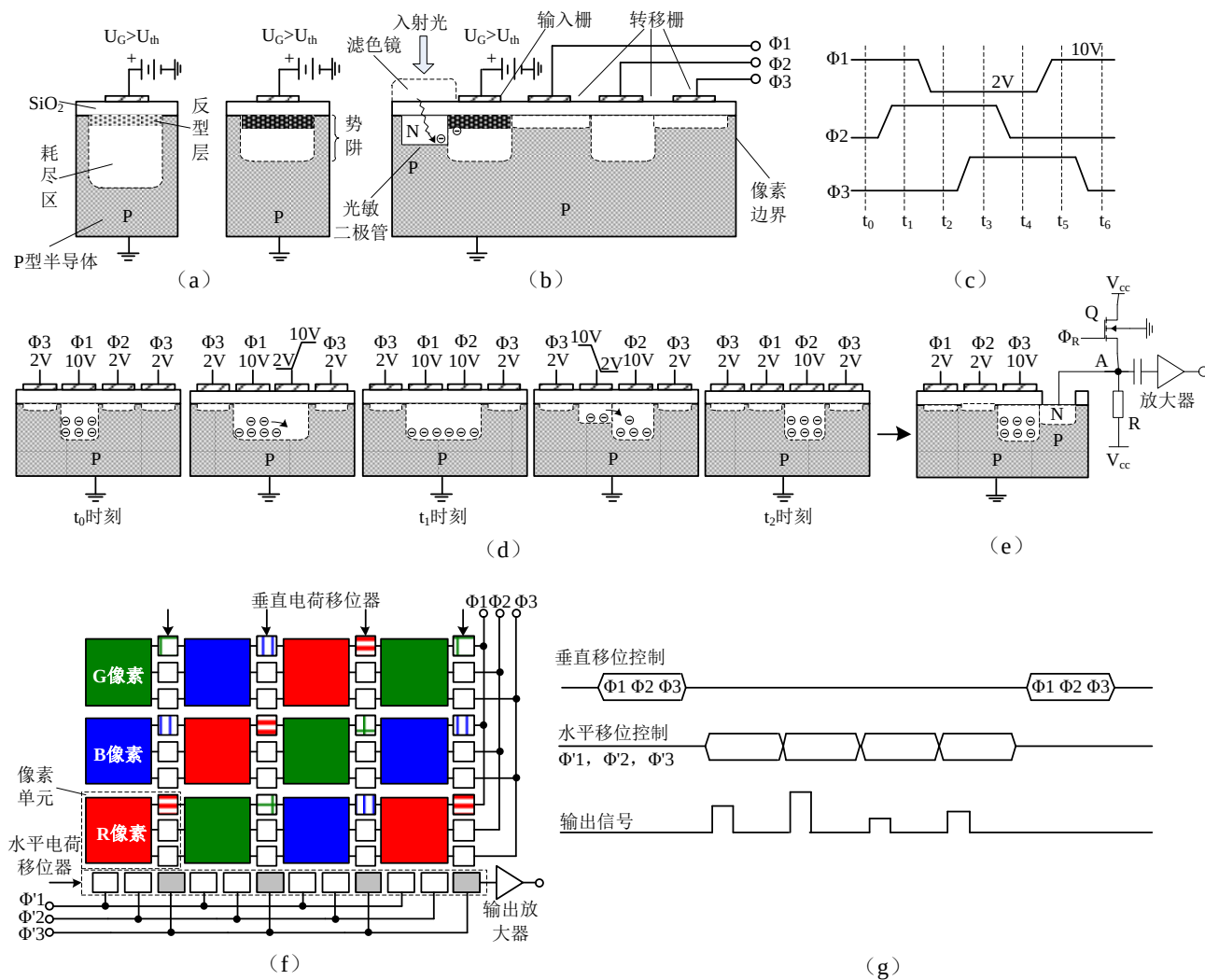
巨磁阻 (GMR) 效应



半导体式色敏传感器：(a)非晶态硅色敏传感器；(b)双色硅色敏传感器

- 无定形硅对可见光范围的灵敏度高，灵敏光谱与人的视觉灵敏度接近
- 无定形硅对红外不敏感
- 同一硅基片两个深浅不同的PN结光电二极管对光谱的响应不同
- PN结的短路电流比值与光的强度无关，而只与光的波长呈单调对应关系

物性型传感器—色敏和图像传感器 (2)



CCD（电荷耦合器）图像传感器的原理：(a)MOS电容

(b)电荷积分成像原理

(c)三相栅控制时序

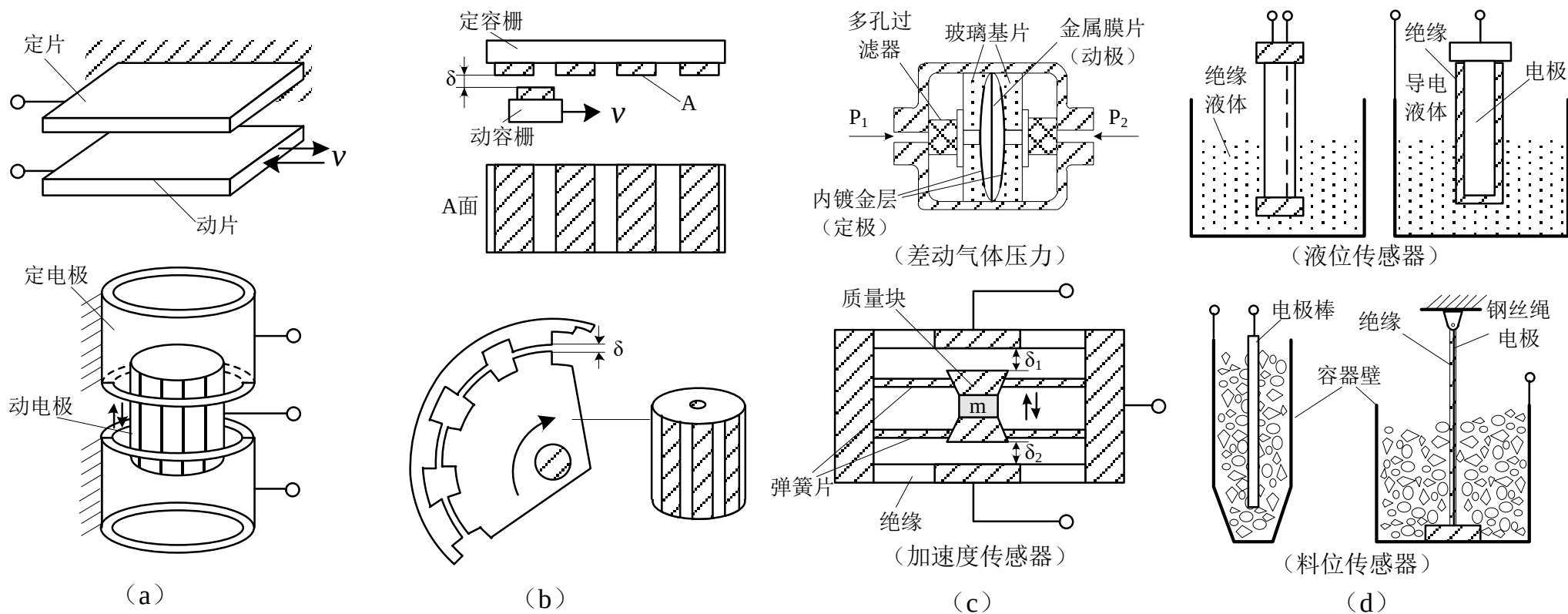
(d)三相栅控制电荷转移原理

(e)电荷检测电路

(f)面CCD阵列结构原理

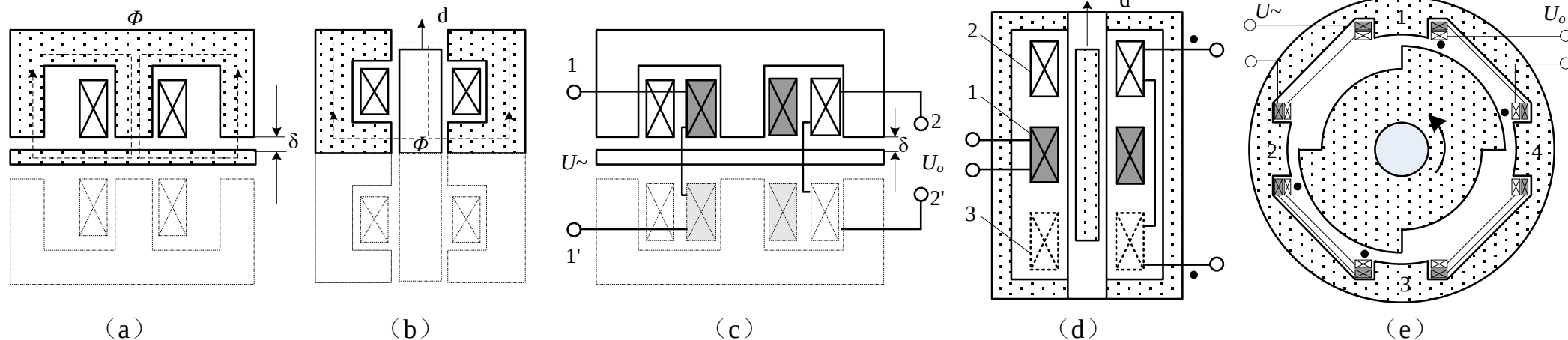
(g)面CCD的控制时序

结构型传感器—电容式传感器



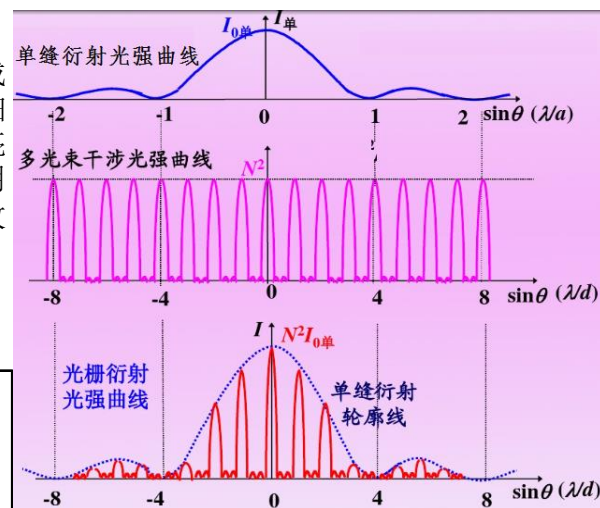
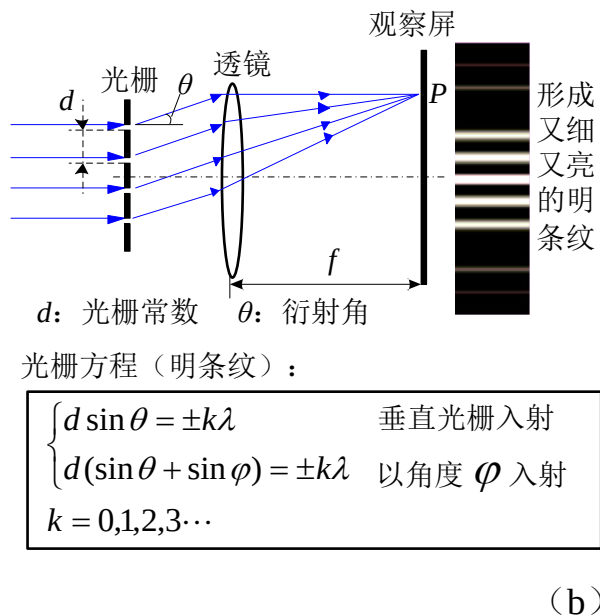
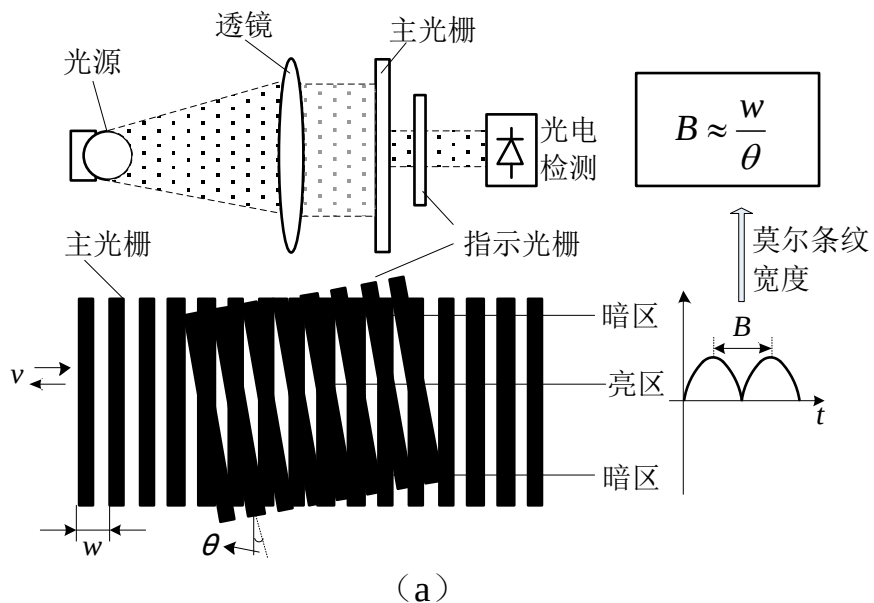
电容传感器：(a)变面积；(b)水平和旋转容栅；(c)变间距；(d)变介电常数

结构型传感器—电感和变压器式传感器



电感和变压器传感器：(a)变气隙电感传感器；(b)变截面电感传感器；(c)变气隙变压器传感器
(d)三节变截面差动变压器；(e)四极测角度差动变压器

结构型传感器—光栅传感器



光栅式传感器: (a)光栅式传感器; (b)衍射光栅





谢谢观看